

Buy late MoE when F/T exists.

ForceVLA keeps your π_0 FM controller and ships 6-axis force through FVLMoE.
Yu*, Liu*, Yu† et al. — arXiv 2505.22159v3. LAST Alt row.

60.5 vs 37.3 (+23.2)

naive F +2.9 only

early MoE → 0%

Don't concat-only. FAST + naive force hurts (31→14.2). Pitch contact-rich specialist augmentation — not a $\pi_{0.5}$ open-world replacement.

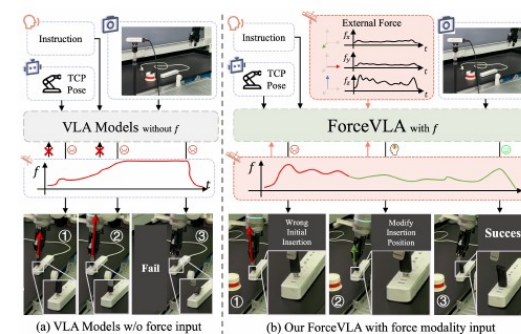


Figure 1: Comparison between ForceVLA and baselines without force input. Without force feedback, the policy fails to correct pose errors and completes insertion incorrectly. In contrast, ForceVLA leverages external force signals to adjust insertion strategies dynamically, leading to successful

Five facts that do not change with the seat

Late 6-axis F/T + FVLMoE

01

After VLM (PaliGemma/SigLIP): project wrench → force token → FVLMoE (E=4, top-1) → additive guide into π_0 flow-matching action head. Force stays late.

Contact-rich on Flexiv

02

Insert USB / plug, pump bottle, wipe board, peel cucumber. ForceVLA-Data: 244 traj / ~140k steps / ~50 demos per task. Quest3 teleop.

Must-cites

03

Avg 60.5 vs π_0 -base w/o F 37.3 (+23.2). Naive w/ F only 40.2 (+2.9). Early MoE-before-VLM 0% → late FVLMoE 80% (Tab.3). Occlusion 90%. FAST+naive force 31.0→14.2.

CLEAR contrasts

04

NOT MoPA-PD ancestor (MoPA = MP privilege→vision student; ForceVLA = deploy force into VLA). NOT JEPA/WM child. No $\pi_{0.5}$ bake-off — do not invent.

Number hygiene

05

Typo fix: 37.3 = w/o F (not w/ F). Preserve paper numbers. No invented BEHAVIOR SR. Pump caveat: naive w/F 83 > ForceVLA 67 — do not overclaim.

After MoPA-PD, lock the deploy-side force answer

MoPA was train privilege

Prior Alt row: classical MP at train $\rightarrow$ vision student. ForceVLA is the complementary arrow: add a sensor modality to a deployed VLA.

Vision-only π_0 still fails contact

Misaligned insert, occlusion, surface force. Semantic VLAs inherit open-world strength and still jam on wrench-dominated phases.

No new foundation pretrain

Keep PaliGemma + FM action expert. Productize F/T + late MoE. HIGH force/contact flag without restarting pretrain.

Closes the Alt column

Sheet ends with force-in-VLA before pure open-world weeks. Stakeholder needs a buy/walk, not a genealogy lecture.

Late force. Sparse MoE. FM head stays.

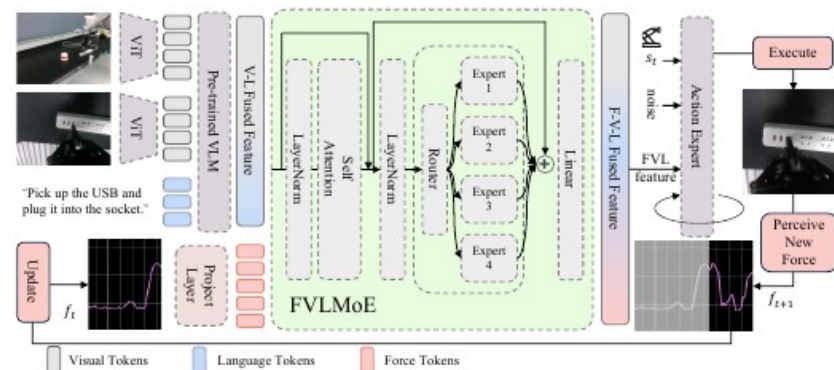


Figure 3: Overview of our ForceVLA model. Visual and language inputs are processed by a pre-trained VLM to form contextual embeddings. External force signals are projected and fused with VLM outputs via the FVLMoE module. The resulting multimodal features guide a flow-based action head to generate contact-aware robot actions.

that are used to select a small subset of k experts (typically $k = 1$ or $k = 2$, where $k \ll N$) from the total pool of N experts. The input token x is then routed only to these k active experts. The outputs of these active experts, $E_i(x)$, are subsequently aggregated, commonly through a weighted sum where the weights $g_i(x)$ are also derived from the gating network. The final output $y(x)$ of the MoE layer can be expressed as: $y(x) = \sum_{i \in \text{TopK}(G(x))} g_i(x) E_i(x)$, where $\text{TopK}(G(x))$ denotes the set of indices of the top- k experts selected by the gating network for input x .

4 ForceVLA

After VLM

Force token lands post PaliGemma. Pretrained VL geometry preserved.

E=4 top-1

FVLMoE routes VL+force; residual $\rightarrow 1024 \rightarrow$ additive FM suffix.

Not FAST

ForceVLA keeps flow matching. π_0 -fast is a baseline foil only.

Buy late MoE when F/T exists. Walk concat-only.

BUY THE PATTERN

Reliable 6-axis F/T on the arm

Integrated or retrofit. Tasks = insert / wipe / press / peel-like.

Already shipping π_0 / OpenPI FM

Keep the controller. Add late force token + FVLMoE. Collect sync F/T demos (~50/task here).

Contact is the failure mode

Vision-only stacks jam on alignment / occlusion / compliance. That is the buy.

WALK / DO NOT BOLT

No F/T hardware

Pattern blocked. Estimated wrench / tactile skins are research paths (paper limitation).

Just concat force to state

Paper: +2.9 pts only (37.3→40.2). Budget the MoE / late-fusion engineering.

FAST-only compact policies

Naive force hurts π_0 -fast 31.0→14.2. Don't bolt F/T without capacity / pretrain story.

Fusion quality beats force presence

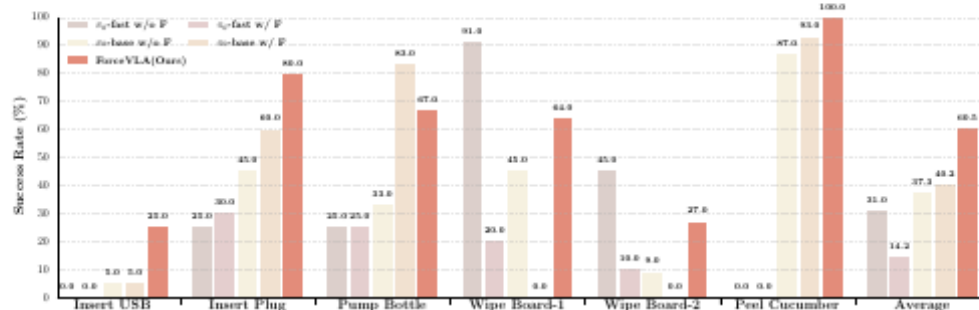


Figure 5: Main task success rates across different methods. ForceVLA significantly outperforms all baselines on five contact-rich tasks. Incorporating external force feedback improves performance for π_0 -base model, while our method achieves the highest average success rate, demonstrating robust performance under complex interaction dynamics. “Wipe Board-1” indicates the success rate of successfully performing the wiping motion, while “Wipe Board-2” refers to the success rate of completely erasing the markings.

ForceVLA shows an improvement of 23.2%. This highlights the substantial benefit of incorporating and effectively processing multimodal information, including force. Table 1 further highlights ForceVLA’s superior performance on the intricate *cucumber peeling* task. Our model excelled on both key metrics: it achieved the longest average peel length per stroke (14.12 cm $\uparrow$), indicating better ability to execute high-fidelity surface manipulation through stable tool orientation, adaptive centering, and sustained surface contact compared to both π_0 -base w/o F (12.17 cm) and π_0 -base w/ F (13.17 cm).

Headline

60.5 vs 37.3 (+23.2)

vs π_0 -base w/o F. Typo fix: 37.3 is w/o F.

Naive concat

40.2 (+2.9 only)

Presence $\neq$ integration.

FAST warning

31.0 $\rightarrow$ 14.2 when force is bolted naively.

Plug headline

ForceVLA 80% on Insert Plug.

Four costs the abstract does not waive

F/T cost + estimated wrench

Paper uses estimated external wrench on integrated Flexiv-class sensing. Calibration and accessibility are stated limitations.

Sync demos are not free

ForceVLA-Data: 244 traj / ~50 per task. OXE alone does not give you wrench-synced contact demos.

USB and unstable socket

Multi-task USB stays at 10%. Unstable Socket: ForceVLA 20% (π_0 -fast w/F best at 30%). Not a universal fix.

Not a $\pi_{0.5}$ replacement

Pitch contact-rich specialist augmentation. Do not brief open-world homes as solved. No $\pi_{0.5}$ numbers in this paper.

Five contact-rich Flexiv tasks — what you are buying for

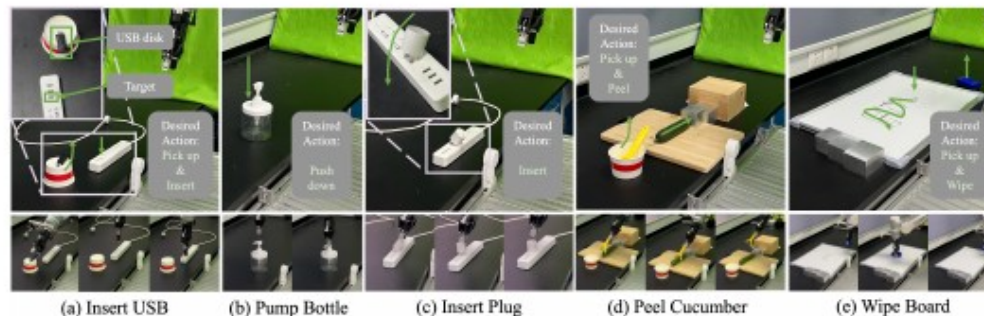


Figure 4: Overview of task setups used in evaluation. (a) Insert USB, (b) pump bottle, (c) insert plug, (d) peel cucumber, and (e) wipe board. These tasks span diverse contact dynamics and manipulation skills, from precise insertions to tool-mediated surface interactions.

5 Experiments

This section presents a comprehensive suite of real-world contact-rich manipulation experiments and analytical studies to empirically validate the ForceVLA model. The evaluation is structured around four core research questions: (1) the overall effectiveness of ForceVLA compared to baselines that

Insert

USB + plug. Alignment + force-controlled seating. Plug is the 80% story; USB stays hard.

Press / wipe

Pump bottle; wipe motion vs full erase. Continuous contact force.

Surface

Cucumber peel: length $\uparrow$ / strokes $\downarrow$. Continuous tool force, not binary insert.

When vision fails, force must carry – Tab.2 90%

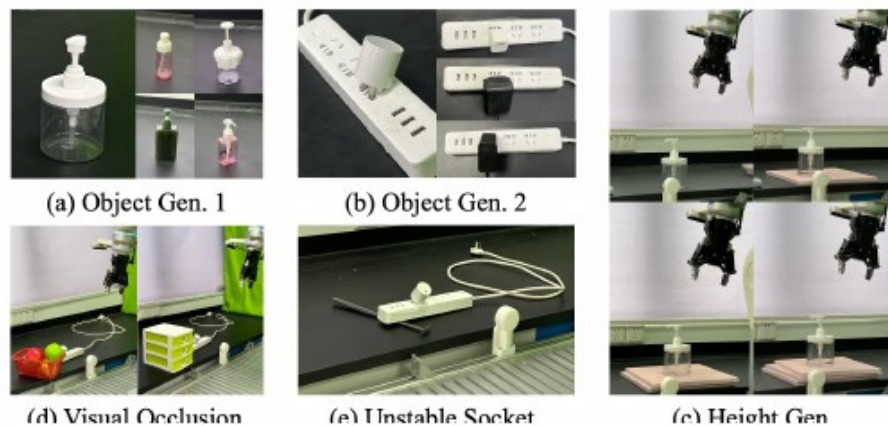


Figure 6: Variants of generalization settings used in our experiments. (a–b) Different object geometries; (c) variation in socket height; (d) partial visual occlusion; (e) unstable socket conditions. These scenarios evaluate robustness under diverse physical and perceptual perturbations.

Table 2: Success rates (%) of different models under various experimental conditions. Maximum values in each column are highlighted in **bold**; second-best values are underlined.

Model	Object Gen. 1	Object Gen. 2	Height Gen.	Visual Occlusion	Unstable Socket	Average
π_0 -base[10] w/o F	48.00%	10.00%	66.67%	60.00%	10.00%	38.93%
π_0 -base[10] w/ F	32.00%	10.00%	77.78%	30.00%	10.00%	31.96%
π_0 -fast[25] w/o F	80.00%	<u>35.00%</u>	88.89%	50.00%	10.00%	52.78%
π_0 -fast[25] w/ F	32.00%	5.00%	44.44%	50.00%	30.00%	32.29%
ForceVLA (Ours)	80.00%	40.00%	88.89%	90.00%	<u>20.00%</u>	63.78%

height and measures success under torque limits. (d) **Visual Occlusion** where parts of the blue and

Visual Occlusion

ForceVLA 90%

π_0 -base w/o F 60%

π_0 -base w/ F 30% (naive hurts)

Gen average

ForceVLA 63.78%

Unstable Socket still weak: ForceVLA 20; fast w/F 30 best.

Buy message

Force substitutes when wrist cam / clutter occludes contact.

Same Alt row as MoPA. Opposite arrow. Not a JEPA child.

vs MoPA-PD

MoPA: privileged classical MP at train $\rightarrow$ distill to vision. ForceVLA: deploy force modality into a VLA. Shared syllabus slot; inverse mechanism.

vs π_0 / FAST

π_0 is parent foil / backbone. π_0 -fast is baseline family. ForceVLA keeps FM. Naive force on FAST hurts.

VS $\pi_{0.5}$

Cited in related work only. No bake-off. Do not invent wins/losses. Open-world language stays $\pi_{0.5}$ -class; force is contact specialist.

vs JEPA / WM

CLEAR contrast poles only. ForceVLA is not a world-model paper. Do not invent ancestry.

What to say in the LAST Alt seat

Schedule

LAST Alt approaches row. Locks force-in-VLA before closing the sheet.

Buy

Late MoE fusion pattern on π_0 FM when F/T exists and contact fails vision-only.

Do not buy

Concat-only. FAST+naive force. Claim that vision-only homes are solved.

One-liner

ForceVLA is modality productization: keep π_0 FM; ship F/T + FVLMoE when contact is the failure mode.

Buy late MoE. Refuse concat-only. No $\pi_{0.5}$ laundry.

ForceVLA = π_0 + late 6D force + FVLMoE → contact-rich Flexiv policy.

Say: 60.5 vs 37.3 (+23.2). Naive +2.9. Early MoE 0%. Occlusion 90%. FAST+F hurts.

Do not say: beats $\pi_{0.5}$; MoPA ancestor; JEPA child; invented BEHAVIOR SR; wins Pump vs naive.

Next: Scientific Reviewer grades late MoE as the real contribution.